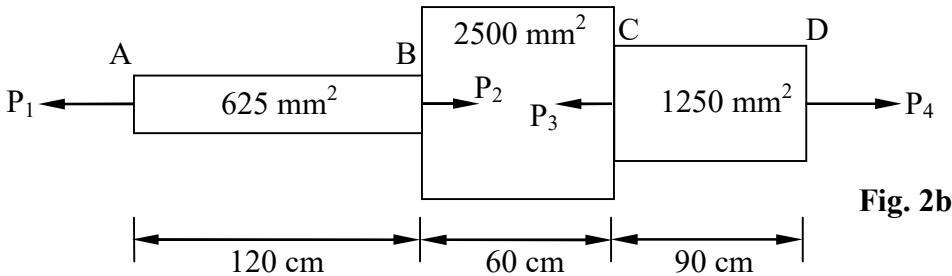
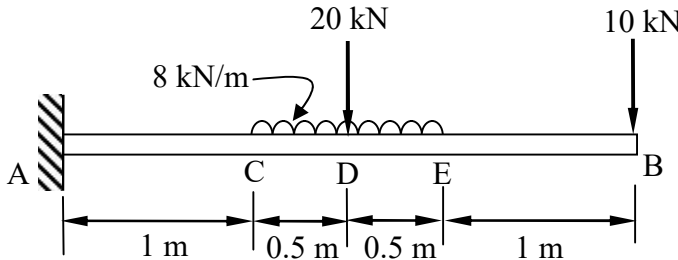
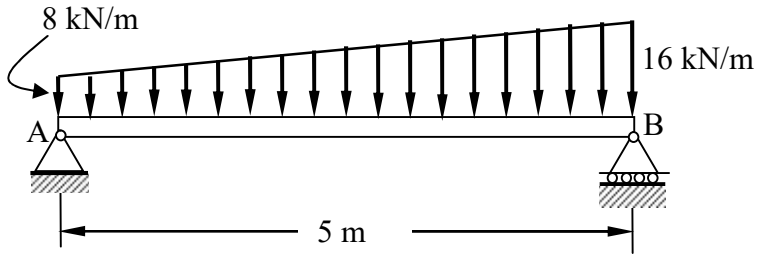
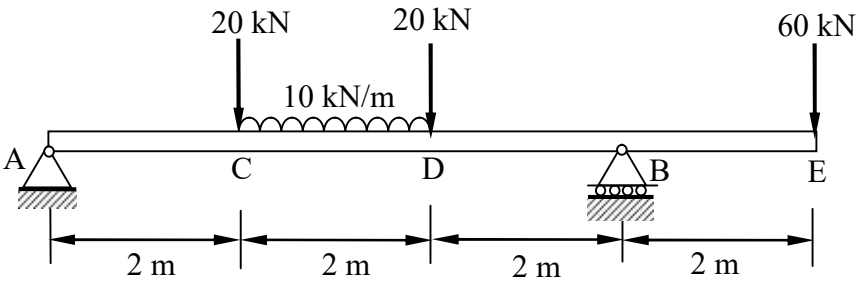
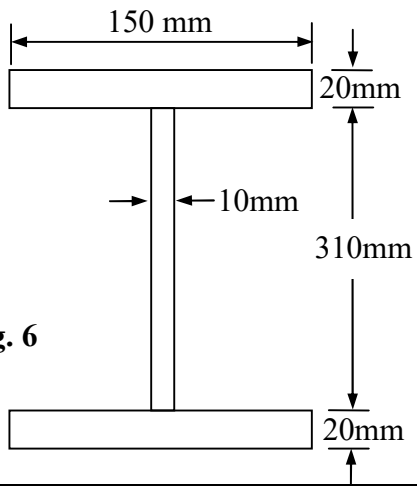
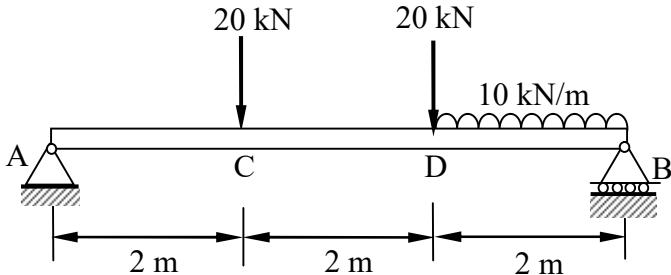


Course Code:B20BS2101																				
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20																
II B. Tech I Semester Regular Examinations																				
NUMERICAL METHODS & VECTOR CALCULUS																				
(Common to CE, CSE, EEE & IT)																				
Time: 3 Hrs.			Max. Marks:70																	
Answer any one Question from Each Unit																				
All questions carry equal Marks																				
	UNIT-I			CO	KL	M														
1.a)	Determine a real root of the equation $x \log_{10} x = 1.2$ by Regula-falsi method and correct to two decimal places			1	K3	7														
b)	Determine the cube root of 41 using Newton-Raphson method			1	K3	7														
(OR)																				
2. a)	Determine Newton's forward difference interpolation formula find Y (3), from the following table <table border="1"><tr><td>X</td><td>0</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>Y</td><td>7</td><td>11</td><td>14</td><td>18</td><td>24</td><td>32</td></tr></table>			X	0	5	10	15	20	25	Y	7	11	14	18	24	32	1	K3	7
X	0	5	10	15	20	25														
Y	7	11	14	18	24	32														
b)	Using Lagrange's interpolation formula find Y (10) from the following table <table border="1"><tr><td>X</td><td>5</td><td>6</td><td>9</td><td>11</td></tr><tr><td>Y</td><td>12</td><td>13</td><td>14</td><td>16</td></tr></table>			X	5	6	9	11	Y	12	13	14	16	1	K3	7				
X	5	6	9	11																
Y	12	13	14	16																
UNIT-II																				
3.a)	Evaluate $\int_0^1 x^3 dx$ with five subintervals by trapezoidal rule			2	K3	7														
b)	Evaluate $\int_0^2 \frac{dx}{x^2+x+1}$ by using Simpsons 1/3 rd rule with $h = 0.25$			2	K3	7														
(OR)																				
4. a)	Employ Taylor's method to obtain approximate value of y at $x=0.2$ for the differential equation $\frac{dy}{dx} = 2y + 3e^x, y(0) = 0$			2	K3	7														
b)	Evaluate $y(0.2)$ using Runge-Kutta 4 th order method, given $\frac{dy}{dx} = \frac{y^2-x^2}{y^2+x^2}, y(0) = 1$.			2	K3	7														
UNIT-III																				
5.a)	Apply change the order of integration and evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dx dy$			3	K3	7														
b)	Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates			3	K3	7														
(OR)																				

6. a)	Evaluate $\int_{-1}^1 \int_0^2 \int_{x-z}^{x+z} (x+y+z) dx dy dz$	4	K3	7
b)	Determine the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$	4	K3	7
UNIT-IV				
7. a)	Obtain the directional derivative of $\phi = xy + yz + zx$ at A in the direction of AB where $A = (1, 2, -1)$, $B = (5, 6, 8)$.	5	K3	7
b)	Determine the values of a and b such that the surface $ax^2 - byz = (a+2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1, -1, 2)$.	5	K3	7
(OR)				
8. a)	Show that the vector $(x^2 - yz)\bar{i} + (y^2 - zx)\bar{j} + (z^2 - xy)\bar{k}$ is irrotational and find its scalar potential.	5	K3	7
b)	Determine $\text{Curl } \vec{F}$ and $\text{div } \vec{F}$ for $\vec{F} = x^2y\bar{i} - 2xz\bar{j} + 2yz\bar{k}$	5	K3	7
UNIT-V				
9.a)	Determine the work done in moving a particle once round the circle $x^2 + y^2 = 9$ in the xy- plane by the force $\vec{F} = (2x - y - z)\bar{i} + (x + y - z^2)\bar{j} + (3x - 2y + 4z)\bar{k}$	6	K3	7
b)	Evaluate the line integral by Stokes's theorem for the vector function $\vec{F} = y^2\bar{i} + x^2\bar{j} + (z + x)\bar{k}$ and C is the triangle with vertices $(0,0,0)$, $(1,0,0)$ and $(1,1,0)$.	6	K3	7
(OR)				
10.	Verify Green's theorem in the plane For $\oint_C [(3x^2 - 8y^2)dx + (4y - 6xy)dy]$, where C is boundary of the region defined by $y = \sqrt{x}$, $y = x^2$	6	K3	14

Course Code: B20CE2101				
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			R20	
II B.Tech I Semester –MODEL PAPER				
STRENGTH OF MATERIALS				
(For CE)				
TIME: 3Hrs.			Max. Marks: 70M	
Answer any One Question from Each Unit				
All questions carry equal marks				
	UNIT-I	CO	KL	M
1.a)	Explain the Stress-Strain diagram for mild steel.	1	K2	7
b)	A steel rod of 3cm diameter and 5m long is connected to two grips and the rod is maintained at a temperature of 95°C. Determine the stress exerted when the temperature falls to 30°C if i) the ends do not yield ii) the ends yield by 0.12 cm Take $E= 2\times 10^5 \text{ MN/m}^2$ and $\alpha= 12\times 10^{-6}/^\circ\text{C}$	1	K3	7
(OR)				
2. a)	A bar of 20 mm diameter is tested in tension. It is observed that when a load of 37.7 kN is applied, the extension measured over a gauge length of 200 mm is 0.12 mm and contraction in diameter is 0.0036 mm. Find the Poisson's ratio and elastic constants E, G and K.	1	K3	7
b)	A member ABCD is subjected to point loads P_1 , P_2 , P_3 and P_4 as shown in fig. 2b. Calculate the force P_2 necessary for equilibrium if $P_1 = 45\text{kN}$, $P_3= 450 \text{ kN}$ and $P_4 = 130 \text{ kN}$. Determine the total elongation of the member, assuming the modulus of elasticity to be 210 GN/m^2 . 	1	K3	7
UNIT-II				
3.a)	Draw shear force and bending moment diagrams for the beam shown in Fig. 3a 	2	K3	7
Fig. 3a				

b)	Draw shear force and bending moment diagrams for the beam shown in Fig. 3b  Fig. 3b	2	K3	7
(OR)				
4. a)	Derive the relationship between Load, Shear Force and Bending Moment	2	K3	7
b)	Draw shear force and bending moment diagrams for the beam shown in Fig. 4b  Fig. 4b	2	K3	7
UNIT-III				
5.a)	Derive the Bending Equation.	3	K3	8
b)	A rectangular beam 200mm deep and 300mm wide is Simply Supported over a span of 8m. What uniformly distributed load per metre, the beam can carry, if the bending Stress is not to exceed 120N/mm^2 .	3	K3	6
(OR)				
6.	A I-Section beam 350mmx150mm has a web thickness of 10mm and a flange thickness of 20mm. If the Shear force acting on the section is 40kN. Find the maximum shear stress developed in the I-Section. Draw the shear stress distribution diagram across the section.  Fig. 6	3	K3	14

	UNIT-IV			
7.	A beam AB of 6 m span is simply supported at the ends and is loaded as shown in Fig.7. Find the position and magnitude of maximum deflection using Macaulay's method. Take $E = 2 \times 10^5 \text{ N/mm}^2$, $I = 60 \times 10^6 \text{ mm}^4$.  Fig. 7	4	K3	14
	(OR)			
8. a)	Derive an expression for the Normal and Tangential Stresses on an oblique plane, when the body is subjected to direct stresses in two mutually perpendicular direction accompanied by a shear stress.	4	K3	7
b)	At a point within a body is subjected to two mutually perpendicular directions, the stresses are 150 N/mm^2 (tensile) and 100 N/mm^2 (compression). Each of the above stresses is accompanied by a shear stress of 100 N/mm^2 . Determine the Normal, Shear and Resultant stresses on an oblique plane inclined at 45° with the axis of minor tensile stress.	4	K3	7
	UNIT-V			
9.a)	Derive an expression for Crippling load when both the ends of the column are hinged.	5	K3	7
b)	A hollow cast iron column 200mm outside diameter and 150mm inside diameter, 8m long has both ends fixed. It is subjected to an axial compressive load. Taking a Factor of Safety=6, Yield Stress= 560 N/mm^2 & $\alpha=1/1600$. Determine Safe Rankine's load.	5	K3	7
	(OR)			
10. a)	Derive an expression for Maximum and minimum stresses developed in a column subjected to eccentric loading.	5	K3	7
b)	A column of timber section $15 \text{ cm} \times 20 \text{ cm}$ is 6 m long, one end being fixed and the other end is free. If the Young's modulus for timber = 17.5 kN/mm^2, determine: i) Euler's Crippling load and ii) Safe load for the column if factor of safety = 3.	5	K3	7

Course Code: B20CE2102																
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R20												
II B.Tech I Semester –MODEL PAPER																
ENVIRONMENTAL ENGINEERING																
(For CE)																
TIME: 3 Hrs.			Max. Marks: 70 M													
Answer any One Question from Each Unit																
All questions carry equal marks																
	UNIT-I	CO	KL	M												
1.a)	Explain the physical and chemical characteristics of water.	1	K2	7												
b)	Explain the factors affecting per capita water demand.	1	K2	7												
(OR)																
2. a)	Explain the WHO guidelines for drinking water.	1	K2	7												
b)	The following is the population data of a city available from past census records. Determine the future population of the city in 2030 by i) Arithmetic Increase method ii) Geometrical Increase method. <table border="1"><tr><td>Year</td><td>1960</td><td>1970</td><td>1980</td><td>1990</td><td>2000</td></tr><tr><td>population</td><td>25000</td><td>28000</td><td>34000</td><td>42000</td><td>47000</td></tr></table>	Year	1960	1970	1980	1990	2000	population	25000	28000	34000	42000	47000	1	K3	7
Year	1960	1970	1980	1990	2000											
population	25000	28000	34000	42000	47000											
UNIT-II																
3.a)	Explain the working of Rapid Gravity filter with neat sketch.	2	K2	7												
b)	Explain the theory of sedimentation.	2	K2	7												
(OR)																
4. a)	Explain the various methods of removing excess iron and manganese from ground water.	2	K2	7												
b)	Explain the reverse osmosis with a neat sketch.	2	K2	7												
UNIT-III																
5. a)	Explain with neat sketches the various types of Intake structures based on sources.	3	K2	7												
b)	Explain the distribution layouts in water supply.	3	K2	7												
OR																
6. a)	Explain the factors affecting the selection of pipe materials for water supply.	3	K2	7												
b)	Explain the functioning of any two valves with neat sketch.	3	K2	7												
UNIT-IV																

7.a)	Explain the physical and chemical characteristics of sewage.	4	K2	7
b)	Explain how to determine BOD by dilution method.	4	K2	7
	OR			
8. a)	Explain house plumbing systems with a neat sketch.	4	K2	7
b)	Explain the operation of a grit chamber with a neat sketch.	4	K2	7
	UNIT-V			
9. a)	Explain activated sludge process with a neat sketch.	5	K2	7
b)	Design a septic tank for 200 users.	5	K3	7
	OR			
10. a)	Explain the working and operation of an oxidation pond.	5	K2	7
b)	Explain sludge drying method with a neat sketch.	5	K2	7

Course Code: B20CE2103				
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			R20	
II B.Tech I Semester –MODEL PAPER				
FLUID MECHANICS				
(For CE)				
TIME: 3Hrs.			Max. Marks: 70M	
Answer any One Question from Each Unit				
All questions carry equal marks				
	UNIT-I	CO	KL	M
1.a)	Explain the phenomenon of Capillarity. Obtain the expression for capillary rise of liquid.	1	K2	7
b)	If 10,000 litres of certain liquid weigh 1329KN. Determine i)Specific weight ii) Mass density iii) Specific Volume iv) Specific gravity.	1	K3	7
(OR)				
2. a)	Define Gauge pressure , Absolute pressure and Atmospheric pressure and give the relation between them.	1	K2	7
b)	Petrol of Specific gravity 0.8 flow upwards through a vertical pipe . A &B are two points in the pipe, B being 0.3m higher than A, connections are led from A & B to a U-tube containing mercury. If the difference of pressure between A & B is 0.18Kgf/cm ² . Find the difference in the mercury level in the differential manometer.	1	K3	7
UNIT-II				
3.a)	In a Two-dimensional incompressible flow, the fluid velocity components are given by u=x-4y& v=-y-4x. Show that velocity potential exist and determine its form. Find also the stream function.	2	K3	7
b)	Compare i) Steady flow & unsteady flow i) One dimensional & two- dimensional flow ii) Stream line & path line	2	K2	7
(OR)				
4. a)	Explain the method to find Meta- Centric height experimentally.	2	K2	7
b)	A block of wood of specific gravity 0.7 floats in water. Determine the meta centric height of the block if itssize is 2mx1mx0.8m.	2	K3	7
UNIT-III				
5.a)	Develop an expression for discharge through rectangular notch.	3	K3	7
b)	A right angled V-notch is used for measuring discharge of oil. An error of 1.5mm was made while measuring the head over notch. Determine % error in discharge.	3	K3	7
OR				

6. a)	State Bernoulli theorem. Evaluate the Bernoulli's equation starting from Euler's equation of motion along a stream line.	3	K2	7
b)	The inlet and throat diameter of a horizontal venturimeter are 30cm and 10cm respectively. The liquid flowing through venturimeter is water. The pressure intensity at inlet is 13.734N/cm^2 , while the vacuum pressure head at throat is 37cm of mercury. Find the rate of flow. Take $C_d = 0.98$.	3	K3	7
UNIT-IV				
7.a)	Develop Hagen Poiseuille equation for laminar flow through a circular pipe.	4	K3	7
b)	An oil of viscosity 0.2N/m^2 and specific gravity 0.85 flow through a circular pipe of diameter 75mm and length 250m. The rate of flow through the pipe is 5lps. Find the pressure drop in a length of 250m and shear stress at the pipe wall.	4	K3	7
OR				
8. a)	Explain Prandtl mixing length theory for turbulent shear stress.	4	K2	7
b)	Determine the wall shearing stress in a pipe of diameter 100mm which carries water . The velocities at the pipe centre & 30mm from the pipe centre are 2m/sec and 1.5m/sec respectively. The flow in pipe is given a turbulent.	4	K3	7
UNIT-V				
9.a)	Develop Darcy – Weisbach equation for head loss through a pipe.	5	K3	7
b)	A crude oil of kinematic viscosity 0.4 stoke is flowing through a pipe of diameter 300mm at the rate of 300lit/sec. Find the head loss due to friction for a length of 50m of the pipe.	5	K2	7
OR				
10. a)	Explain the phenomenon of water hammer. Obtain an expression for rise of pressure when the flowing water in a pipe is brought to rest by closing the valve gradually.	5	K2	7
b)	The difference in water surface levels in two tanks, which are connected by three pipes in series of length 300m, 170m & 210m and of diameter 300mm, 200mm & 400mm respectively, is 12m. Determine the rates of flow of water if co-efficient of friction are 0.0005, 0.0052 & 0.0048 respectively neglecting the minor losses.	5	K3	7

Course Code: B20CE2104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
II B.Tech I Semester –MODEL PAPER					
SURVEYING					
(For CE)					
TIME: 3Hrs.			Max. Marks: 70M		
Answer any One Question from Each Unit					
All questions carry equal marks					
			CO	KL	M
UNIT - I					
1.	a).	Explain errors in chaining?	1	K2	7
	b).	Explain different types of tape corrections?	1	K2	7
OR					
2.	a).	Outline differences between magnetic dip and magnetic declination.	1	K2	7
	b).	The bearings of the lines of a closed traverse are 290° 30"; 50° 30'; 196° 0'; 175°30"; 112° 0"; 30° 0"; Determine the included angles and the angular error.	1	K3	7
UNIT - II					
3.	a).	The following consecutive readings were taken with a level and 5 meter leveling staff a continuously sloping ground on a common interval of 20 meters. 0.385 ; 1.030 ; 1.925 ; 2.825 ; 3.730 ; 4.685 ; 0.625 ; 2.005 ; 3.1101 ; 4.485 the R.L of the first point was 208.125 m. Rule out a page of level book and enter the readings. Determine the R.L of the points by rise and fall method.	2	K3	14
OR					
4.	a).	Explain different sources of errors in leveling? How are they eliminated?	2	K2	7
	b).	Explain different Characteristics of contour?	2	K2	7
UNIT - III					
5.	a).	What are the various permanent adjustments of Theodolite? Explain in detail.	3	K2	14
OR					
6.		The bearing of one side of a regular pentagon was found to be N300E. Find bearings of other lines. The following angles were observed in clockwise direction in an open traverse angle ABC = 124°15', angle BCD = 156°30' angle CDE = 102°0' angle DEF = 95°15' angle EFG = 215°30' magnetic bearing of line AB was 241°30'.Determine the bearing of line FG =?	3	K3	14
UNIT - IV					
7.	a).	Define curve? Explain different types of horizontal circular curves.	4	K2	7
	b).	Explain compound curve? With Neat sketches?	4	K2	7
OR					
8.	a).	Explain the advantages of Tachometric surveying.	4	K2	7

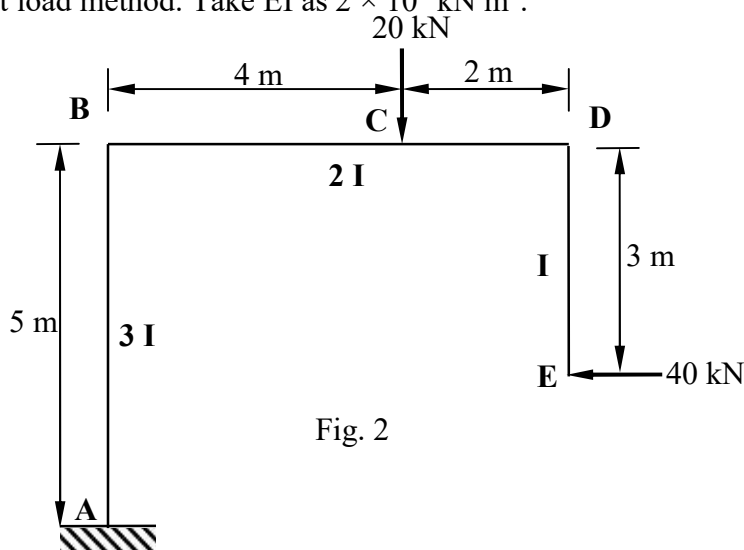
	b).	Explain the conditions in Trigonometrical levelling when base is accessible and inaccessible.	4	K2	7
UNIT - V					
9.	a).	Explain about the different types of EDM instruments.	5	K2	7
	b).	Describe about total station and state its advantage over other methods of Surveying.	5	K2	7
OR					
10.	a).	Explain uses and applications of GPS	5	K2	7
	b).	Explain photogrammetric surveying? Write a note on applications and the Limitations of photogrammetric survey?	5	K2	7

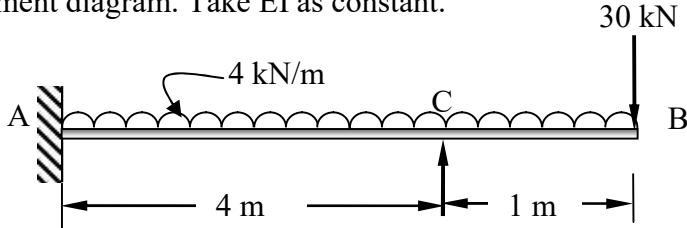
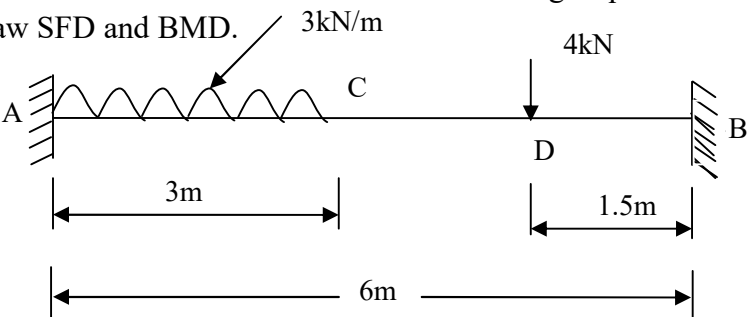
Course Code:B20BS2204					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R 20
II B. Tech II Semester Regular Examinations					
COMPLEX VARIABLES AND STATISTICAL METHODS					
(Common to CE & EEE)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
Q. No.		Question	CO	KL	M
1	A	Determine p such that the function $f(z) = \frac{1}{2} \log_e(x^2 + y^2) + i \tan^{-1}(\frac{yx}{x^2 + y^2})$ will be an analytic function.	1	K3	7
	B	In an electrostatic field, if the potential function is $\phi = 3x^2y - y^3$, then determine the flux function and the complex potential function.	1	K3	7
OR					
2	A	If $f(z) = u + iv$ is an analytic function of $z = x + iy$, establish that $\left[\frac{\partial}{\partial x} f(z) \right]^2 + \left[\frac{\partial}{\partial y} f(z) \right]^2 = f'(z) ^2$.	1	K3	7
	B	Determine the bilinear transformation which maps the points $z= 1, i,-1$ into the points $w = 0, 1, \infty$ respectively. Determine also the fixed points of the transformation.	1	K3	7
3	A	Evaluate $\oint_C \frac{z^3-2z+1}{(z-i)^2} dz$ where C is $ z =2$, using Cauchy integral formula.	2	K3	7
	B	Develop the function $f(z)=\frac{4z+3}{z(z-3)(z-2)}$ as Laurent series (i) (i) in $ z <1$ and (ii) in the annular region $1 < z < 3$.	2	K3	7
OR					
4	A	Determine the residues of $f(z) = \frac{z^3}{(z-1)(z-2)(z-3)}$ at its poles and hence evaluate $\oint_C f(z)dz$, where C is the circle $ z = 2.5$	2	K3	7
	B	Apply the calculus of Residues to evaluate $\int_0^{2\pi} \frac{d\theta}{5-3 \cos \theta}$.	2	K3	7
5	A	Determine the difference equation generated by $y_n = (A + Bn)3^n$.	3	K3	7
	B	Solve the difference equation $y_{n+2} + y_{n+1} - 56y_n = 2^n(n^2 - 3)$.	3	K3	7
OR					

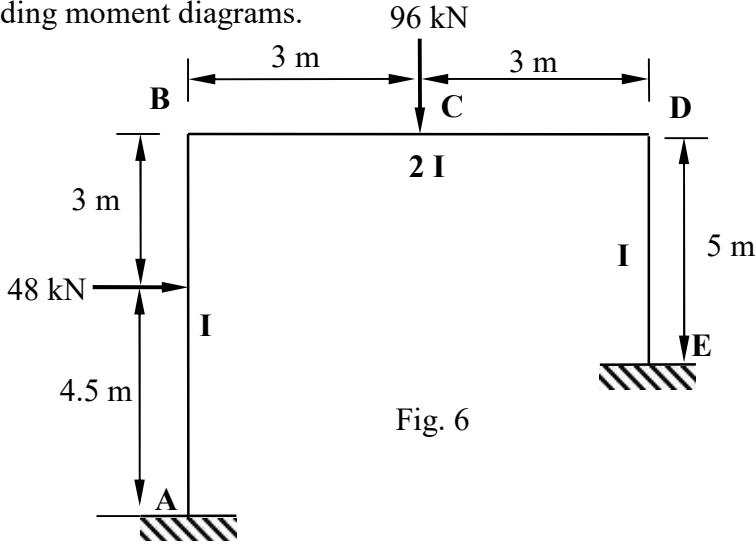
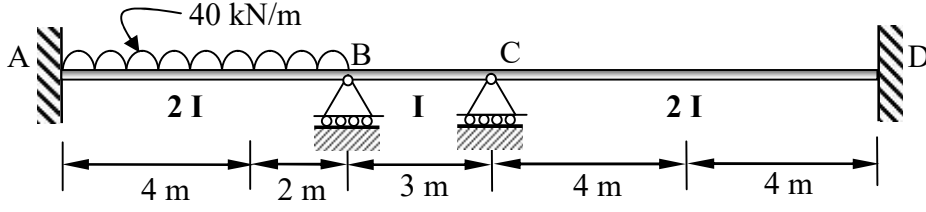
6	A	Given $Z\{u_n\} = \frac{z}{z-1} + \frac{z}{z^2+1}$ determine the Z-transform of u_{n+2} .	4	K3	7																
	B	Utilize Z-transforms to solve $u_{n+2} - 2u_{n+1} + u_n = 3n + 5$.	4	K3	7																
7	A	If X is the random variable of a Poisson distribution such that the probability for X = 2 is two-thirds of the probability for X = 1. Determine the probability for X = 0 and the probability for X = 3. What is the probability for X >3.	5	K3	7																
	B	The average and S.D. of the marks obtained by 500 students in a examination are respectively 40% and 10%. Assuming the normality of the distribution, determine approximately (i)how many will pass if 50% is fixed as minimum, (ii)how many have scored marks above 60%?	5	K3	7																
OR																					
8	A	Derive moment generating function of Poisson distribution.	5	K3	7																
	B	In a Normal distribution, 31% of the items are under 45 and 8% are over 64. Determine the mean and standard deviation of the distribution.	5	K3	7																
9	A	Develop the test procedure for testing the difference of proportions.	6	K3	7																
	B	A sample of 1000 days is taken from meteorological records of a certain district and 120 of them are found to be foggy. Determine the probable limits for the percentage of foggy days in the district.	6	K3	7																
OR																					
10	A	A machine is supposed to produce washers of mean thickness 0.12cm. But the mean thickness of a random sample of 10 washers produced by the machine was found to be 0.128cm with a standard deviation of 0.008cm. Determine whether the machine is working properly at 5% level of significance.	6	K3	7																
	B	The number of aircraft accidents that occurred during the various days of the week is given below: <table><tr><td>Day:</td><td>Sun</td><td>Mon</td><td>Tue</td><td>Wed</td><td>Thu</td><td>Fri</td><td>Sat</td></tr><tr><td>No. of accidents</td><td>14</td><td>16</td><td>8</td><td>12</td><td>11</td><td>9</td><td>14</td></tr></table>	Day:	Sun	Mon	Tue	Wed	Thu	Fri	Sat	No. of accidents	14	16	8	12	11	9	14	6	K3	7
		Day:	Sun	Mon	Tue	Wed	Thu	Fri	Sat												
No. of accidents	14	16	8	12	11	9	14														
Determine whether the accidents are uniformly distributed over the week.																					

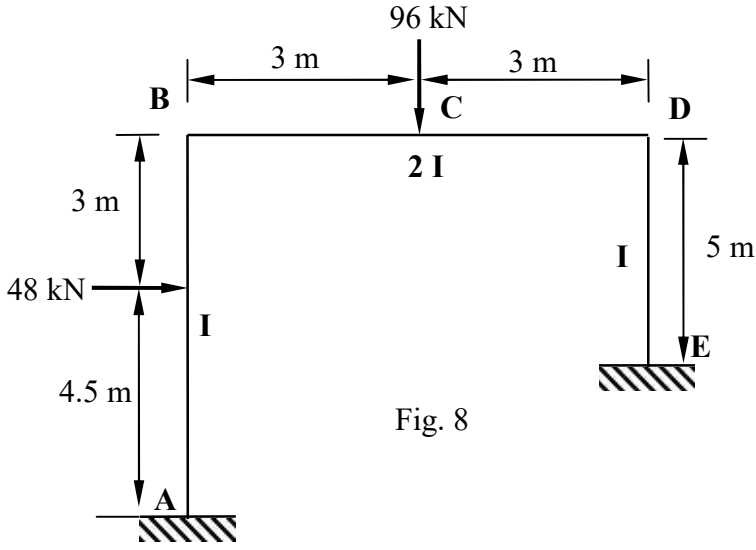
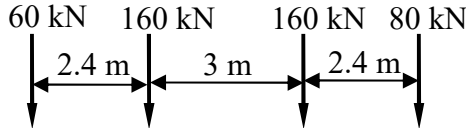
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				Code: B20CE2201		
II B.Tech II Semester –MODEL PAPER				R20		
HIGHWAY ENGINEERING						
(For CE)						
TIME: 3Hrs.				Max. Marks: 70M		
Answer any One Question from Each Unit						
All questions carry equal marks						
			CO	KL	M	
UNIT – I						
1.	(a)	Briefly outline the highway development in India.	1	K2	7	
	(b)	Compare Nagpur & Bombay Road Development plans.	1	K2	7	
OR						
2.	(a)	What is master plan? Explain the importance of master plan in highway planning.	1	K2	7	
	(b)	Briefly outline about rural road development plan?	1	K2	7	
UNIT – II						
3.	(a)	Discuss the design factors of Highway alignment?	2	K3	7	
	(b)	Calculate the safe stopping distance for design speed of 50 kmph for two-way traffic on a two lane road. Assume coefficient of friction as 0.35 and reaction time of driver as 2.5 seconds.	2	K3	7	
OR						
4.	(a)	Derive an expression for extra widening on horizontal curvers.	2	K3	7	
	(b)	Find out the length of transition curve for the following data: Radius of horizontal curve = 400m, design speed = 100 kmph, length of wheel base = 6.2m, number of lanes = 2, rainfall at the location = heavy, terrain condition = hilly, superelevation is introduced by rotating the edges with reference to center line and rate of introduction of superelevation is 1 in 150. Width of highway is 7m.	2	K3	7	
UNIT – III						
5.	(a)	Explain how the speed and delay studies are carried out. What are the various uses of delay studies?	3	K3	7	
	(b)	What are the various types of parking facilities designed for traffic needs?	3	K3	7	
OR						
6.	(a)	Explain the relationship between speed, travel, time, volume, density and capacity.	3	K3	7	
	(b)	Explain the various measures that may be taken to prevent accidents.	3	K3	7	
UNIT – IV						
7.	(a)	Explain the test procedure of shape test for aggregates.	4	K3	7	
	(b)	Estimate the thickness of cement concrete pavement using the method suggested by IRC. Modulus of elasticity of concrete = 3.5 X 10 ⁵ kg/cm ² , Modulus of rupture of concrete = 30 kg/cm ² , Poisson’s ratio of concrete = 0.17, Modulus of subgrade reaction = 5 kg/cm ² , Wheel load = 4800 kg, Radius of contact area = 13 cm.	4	K3	7	
OR						
8.	(a)	Discuss the properties of bitumen.	4	K3	7	
	(b)	Explain group index method of pavement design. What are the limitations of the method?	4	K3	7	
UNIT – V						
9.	(a)	Write a descriptive note on maintenance of highways.	5	K3	7	

	(b)	Write a short note on alligator and reflection cracking.	5	K3	7
OR					
10.	(a)	What are the requirements of material, plants & equipment's for bituminous pavement construction? Discuss briefly.	5	K3	7
	(b)	Give a descriptive note on Pavement Evaluation Techniques.	5	K3	7

Course Code: B20CE2202				
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			R20	
II B.Tech II Semester –MODEL PAPER				
STRUCTURAL ANALYSIS				
(For CE)				
TIME: 3 Hrs.			Max. Marks: 70 M	
Answer any One Question from Each Unit				
All questions carry equal marks				
	UNIT-I	CO	KL	M
1.a)	A beam of constant cross section 4m long is free supported at its ends. It is loaded at points 1m from each end with a load of 2kN. Find the ratio of the deflections in the centre of the beam to the deflection at the point under one of the loads. Use conjugate beam method.	1	K3	7
b)	A tension bar 5m is made up of two parts 3m of its length has cross sectional area of 10cm ² while the remaining 2m has a cross sectional area of 20 cm ² . An axial load of 80 kN is gradually applied. Find the total strain energy produced in bar and compare this value with that obtained in a uniform bar of same length and having the same volume, when under the same load. Take E = 2 x 10 ⁵ N/mm ² .	1	K3	7
(OR)				
2.	Determine the horizontal deflections at ‘E’ in the frame shown in the fig. 2 using unit load method. Take EI as 2 × 10 ⁴ kN m ² .  Fig. 2	1	K3	14

UNIT-II				
3.	Analyse the propped cantilever beam shown in the figure 3 and draw the bending moment diagram. Take EI as constant.  Fig. 3	2	K3	14
(OR)				
4.	A beam AB of uniform section and 6m span is built in at the ends. A uniformly distributed load of 3 kN/m runs over the left half of the span and in addition there is a concentrated load of 4kN at right quarter as shown in fig. 4. Draw SFD and BMD.  Fig.4	2	K3	14
UNIT-III				
5.	A beam ABCD is continuous over three spans AB = 8m, BC = 4m and CD = 8m. The beam AB and BC is subjected to a uniformly distributed load of 1.5kN/m, whereas there is a central point load of 4kN in CD. The moment of inertia of AB and CD is 2I and that of BC is I. The end A and D are fixed. During loading the support A sinks down by 10 mm. Determine the fixed end moments by Theorem of three moments method and draw BMD. Take $E = 200\text{Gpa}$ and $I = 16 \times 10^6 \text{mm}^4$.	3	K3	14
(OR)				

6.	Analyse the portal frame shown in fig. 6 using force method and draw shear force and bending moment diagrams.  Fig. 6	3	K3	14
UNIT-IV				
7.	Analyse the continuous beam shown in Figure 7 by Slope deflection method, if the support 'C' sinks by 10 mm. Take $E = 200 \text{ GPa}$ and $I = 6 \times 10^8 \text{ mm}^4$. Sketch the bending moment diagram.  Fig. 7	4	K3	14
(OR)				

8.	Analyse the portal frame shown in fig. 8 using slope deflection method and draw shear force and bending moment diagrams.  Fig. 8	4	K3	14
UNIT-V				
9.	A uniformly distributed load of 90 kN/m and 6 m long rolls across a simply supported beam of 16 m long. Calculate the maximum shear force and maximum bending moment at a section 4 m from the left-hand support and at a section 10 m from the left-hand support using influence line diagrams. Also determine absolute maximum bending moment.	5	K3	14
(OR)				
10.	A system of 4 loads separated by a specified distance as shown in fig 10 crosses a simply supported beam of 18 m span from left to right. Determine the absolute maximum bending moment on the beam and maximum shear force at center of the beam using ILD's.  Fig. 10	5	K3	14

Course Code: B20CE2203				
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			R20	
II B.Tech II Semester –MODEL PAPER				
HYDRAULICS AND HYDRAULIC MACHINERY				
(For CE)				
TIME: 3Hrs.			Max. Marks: 70M	
Answer any One Question from Each Unit				
All questions carry equal marks				
	UNIT-I	CO	KL	M
1.a)	Explain any two Dimensionless numbers.	1	K2	5
b)	Using Buckingham's π theorem, show that velocity through a circular orifice is given by $V=\sqrt{(2Gh)\phi[D/H, \mu/\rho VH]}$ where H is head causing flow, D is diameter of orifice, μ is coefficient of viscosity, ρ is mass density and g is acceleration due to gravity.	1	K3	9
(OR)				
2. a)	Explain separation of boundary layer with a neat figure.	1	K2	7
b)	A Thin plate is moving in still atmospheric air at a velocity of 5m/sec. The length of plate is 0.6m and width 0.5m. Determine i) thickness of boundary layer at the end of the plate ii) drag force on one side of the plate. Take density of air is 1.24kg/m ³ and kinematic viscosity 0.15 stokes.	1	K3	7
UNIT-II				
3.a)	Develop an expression for Drag and Lift for a stationary solid body.	2	K3	7
b)	A flat plate 1.5m x 1.5m moves at 50km/hr in stationary air of density 1.15Kg/m ³ . If Coefficient of drag and lift are 0.15 and 0.75 respectively. Determine Lift force, drag force and resultant force.	2	K3	7
(OR)				
4. a)	Develop an expression for the force exerted on flat vertical plate moving in the direction of the jet.	2	K3	7
b)	Water is flowing through a pipe at the end of which a nozzle is fitted. The diameter of the nozzle is 100mm and the head of water at the centre nozzle is 100m. Find the force exerted by the jet of water on a fixed vertical plate. The Coefficient of velocity is given as 0.95.	2	K3	7

	UNIT-III			
5.a)	Compare Impulse and Reaction Turbines.	3	K3	7
b)	A pelton wheel is to be designed for a head of 60m when running at 200rpm. The peltonwheel develop 95.6475KW shaft power. The velocity of the bucket=0.45 time the velocityof the jet, overall efficiency= 0.85 and coefficient of velocity is equal to 0.98.	3	K3	7
	OR			
6. a)	Develop an expression for specific speed of the turbine.	3	K3	7
b)	A turbine develops 9000KW when running at 10rpm. The head on the turbine is 30m. If the head on the turbine is reduced to 18m, determine the speed and power developed by the turbine.	3	K3	7
	UNIT-IV			
7.a)	Explain Net Positive Suction Head (NPSH)in Centrifugal Pump.	4	K2	7
b)	A centrifugal pump delivers water against a net head of 14.5m and a design speed of1000rpm. The vanes are curved back to an angle of 30 degrees with periphery. The impeller diameter is 300mm and outlet width is 50mm. Determine the discharge of the pump if manometric efficiency is 95%.	4	K3	7
	OR			
8. a)	Explain the working principle of Reciprocating Pump with a neat figure.	4	K2	7
b)	A single acting reciprocating pump running at 50rpm, delivers 0.01m ³ /sec of water. The diameter of the piston is 200mm and stroke length is 400mm. Determine i) Theoretical discharge ii) coefficient of discharge iii) Slip and percentage slip of the pump.	4	K3	7
	UNIT-V			
9.a)	Evaluate the condition for the best side slope of the most economical trapezoidal section.	5	K3	7
b)	A rectangular channel carries water at the rate of 400lit/sec, when the bed slopeis1 in 2000. Find the most economical dimensions of the channel if C=50.	5	K3	7
	OR			
10. a)	Develop an expression for the depth of hydraulic jump in terms of upstream Froude number.	5	K3	7
b)	The specific energy for a 5m wide rectangular channel is to be 4Nm/N. If the rate of flow of water through the channel is 20m ³ /sec, determine the alternate depths of flow.	5	K3	7

Course Code:B20CE2204					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R20	
II B.Tech II Semester –MODEL PAPER					
DESIGN OF REINFORCED CONCRETE STRUCTURES					
(For CE)					
TIME: 3Hrs.			Max. Marks: 70M		
Answer any One Question from Each Unit					
All questions carry equal marks					
			CO	KL	M
UNIT – I					
1	Determine the ultimate moment of resistance of a rectangular beam section 250 mm wide and 500 mm overall depth. When it is reinforced with 2 numbers of 16 mm diameter bars on compression side and 4 numbers of 25 mm diameter bars on tension side both at an effective cover of 40 mm. Use M20 grade of concrete and Fe415 grade of steel.	1	K3	14	
OR					
2	A T- beam floor consists of 125 mm thick RC slab with 300 mm wide beam . The beams are spaced at 4 m center to center and their effective span is 7 m . If the super imposed load of 6 kN / m , design an intermediate beam . The beams are supported on 230 mm thick wall . Take M25 grade of concrete and Fe415 grade of steel .	1	K3	14	
UNIT – II					
3	A simply supported RCC beam 250 mm X 450 mm (effective) is reinforced with 4 bars of 18 mm diameter bars . Design the shear reinforcement if M20 2 grade of concrete and Fe415 steel is used and beam is subjected to shear force of 150 kN at service state .	2	K3	14	
OR					
4	Design a rectangular beam of size 300 mm x 600 mm subjected to a factored bending moment of 150 kN - m , factored shear force of 100 kN and factored torsional moment of 50 kN - m . Take M20 grade of concrete and Fe415 grade of steel	2	K3	14	
UNIT – III					
5.	Design a slab for a room 4 m x 3.5 m clear in size, if the super imposed load is 3 kN/m2 and floor finish of 1 kN/m2. The edges of the slab are simply supported and corners are not held down. Use M20 grade of concrete and Fe415 grade of steel.	3	K3	14	
OR					
6.	Design a slab for a room 4 m x 3.5 m clear in size, if the super imposed load is 3 kN/m2 and floor finish of 1 kN/m2. The edges of the slab are simply supported and corners are held down. Use M20 grade of concrete and Fe415 grade of steel.	3	K3	14	
UNIT – IV					
7.	Design a circular column of diameter 400 mm is subjected to a load of 1200	4	K3	14	

	kN. The column is having spiral ties. The column is 3 m long and effectively held in position at both but not restrained against rotation. Use M25 grade of concrete and Fe415 steel.			
OR				
8.	Design a square column of size 500 mm x 500 mm carrying an axial factored load of 2000 kN and factored moments, $M_{ux} = 120$ kN-m, $M_{uy} = 160$ kN-m. Assume that the moments due to minimum eccentricities are less than the applied moments. Use M20 grade of concrete and Fe415 grade of steel.	4	K3	14
UNIT – V				
9.	Design a dog legged staircase for an office building in a room measuring 3.0 m X 6.0 m (clear dimensions). Floor to floor height is 3.5 m. The building is a public building liable to overcrowding. Stairs are supported on brick walls 230 mm thick at the end of landing. Use M20 grade of concrete and Fe415 steel.	5	K3	14
OR				
10.	Design a rectangular footing of uniform thickness for an axially loaded column of size 300 mm x 600 mm. The safe bearing capacity of soil is 200 kN/m ² . Load on column is 1150 kN. Use M20 grade of concrete and Fe415 grade of steel.	5	K3	14
